# Priyanshu Sharma

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## Education

**Indian Institute of Technology Madras** Aug. 2024 – May 2026
*Master of Technology in Computer Science Engineering* *CGPA: 8.92*

**Shri G.S. Institute of Technology and Science Indore** Aug. 2019 – May 2023
*Bachelor of Technology in Computer Science and Engineering*

## Experience

**Research Intern** May 2025 – July 2025
*Samsung Research Institute Bangalore* *Bangalore*

- Developed perceptual video preprocessing POC achieving 25% bitrate reduction while maintaining visual quality using Android Studio, Camera2 API, and OpenCV for real-time video processing
- Conducted comprehensive survey and analysis of 15+ preprocessing methodologies, presenting findings to senior members of the team

**Backend Developer Intern** Feb. 2023 – May 2023
*Bynebits Infotech Pvt. Ltd.* *Onsite*

- Designed and developed Django-based websites for organizations, blending powerful backend functionality with sleek frontend frameworks for exceptional user experiences.

**Data Engineer Intern** June 2021 – Aug 2021
*Rivan Solutions Pvt. Ltd.* *Remote*

- Utilized Python libraries (requests, BeautifulSoup, Selenium) to scrape and automate websites, storing data in XLSX files, SQL, and NoSQL databases

## Projects

**Parallel Graph Algorithms & Convolution Optimization** | **GitHub** | *CUDA, C++, Python*

- Implemented CUDA-accelerated Dijkstra's and Borůvka's algorithms, delivering 15x and 12x speedups respectively. Also optimized 2D convolution operations for GPU, achieving 8x speedup over CPU baselines.

**Advanced RNN Architecture Analysis** | **GitHub** | *PyTorch, TensorFlow, Matplotlib*

- Developed end-to-end machine transliteration system using vanilla RNNs, LSTMs, and GRUs with attention mechanisms, achieving 80% character-level accuracy for English-to-Hindi transliteration

**DeepFake Video Detection System** | **GitHub** | *PyTorch, OpenCV, ResNet, LSTM*

- Architected and implemented a Python-based deepfake detection system, integrating cutting-edge technologies such as ResNeXt and LSTM for robust feature extraction and temporal analysis; successfully identified and flagged 90% of manipulated media

**DailyDive: Intelligent News Aggregator** | **GitHub** | *Python, Django, MongoDB, NLP APIs*

- Developed and Designed a real-time news aggregation application, seamlessly integrating content from 5 diverse sources, featuring sentiment-based grouping, bookmarking, and interactive user engagement functionalities

## Technical Skills

**Languages and Frameworks**: Python, C++, JavaScript, SQL, Java, Django, PyTorch, TensorFlow
**Developer Tools**: Android Studio, Git, GitHub Actions, CUDA, Unix/Linux, TCP/IP
**Libraries**: OpenCV, NumPy, Pandas, Matplotlib, Scrapy, BeautifulSoup, PostgreSQL, Camera2 API

## Achievements

**GATE 2024**: All India Rank 2 among 123,967 candidates (99.998th percentile) and AIR 130 in Data Science & AI
**GATE 2023**: All India Rank 139 among 75,679 candidates, consistently maintaining top 0.18% performance for consecutive years